

HF Test Current Injection-Based Self-Sensing Control of PMSM for Low- and Zero-Speed Range Using Two-Degree-of-Freedom Current Control

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Abstract—This paper deals with an innovative self-sensing control scheme based on test current instead of voltage injection. Using two-degree-of-freedom current control, one can overcome the bandwidth limitations that made test current injection unattractive during the last 20 years. The two-degree-of-freedom control scheme consists of a model-based dynamic feedforward control to set the reference response and model reference tracking controllers providing disturbance rejection and position error signals for self-sensing control. Since two-degree-of-freedom control is a model-based control scheme, the compensation for secondary saliencies is inherently included in the precontrol. It is analytically shown that, in the case of alternating test current instead of voltage injection, just the precise knowledge, in general, of the current- and position-dependent secondary diagonal element of the inductance matrix is needed in the case of inductance saliency-based self-sensing control. The effectiveness of the proposed method is proven by experimental results.

Index Terms—Current control, eddy currents, machine vector control, magnetic losses, motion control, permanent-magnet machines, permanent-magnet motors, saliency, self-sensing control, sensorless control, test current injection.

I. INTRODUCTION

FIELD-ORIENTED control of permanent-magnet synchronous machines (PMSMs) without use of a position sensor is still a field of intensive research. Getting rid of the position sensor and related cabling means cost reduction, reduction of installation space, and increased robustness, particularly in harsh environments. A common requirement for industrial applications is that no change of the hardware is allowed. This means no additional sensors may be added. Typically, just the dc bus voltage and the phase currents are available for self-sensing operation.

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The basic idea of self-sensing control is to use the machine itself to estimate the electric rotor position, which is needed for field-oriented control. For low-speed operation, which is addressed in this paper, a sufficient excitation, either by test signal injection or by using pulsewidth-modulated (PWM) inverter switching, is mandatory; otherwise, the rotor position is not observable [1]. It is common to evaluate the high-frequency (HF), typically 500 Hz–2 kHz, behavior of the machine in order to gain the rotor position.

Methods using HF voltage injection and evaluation of the related HF current response are well established in the self-sensing community. The voltage injection methods can be categorized as to the shape of the test signal, i.e., sinusoidal or pulse injection (PWM injection) methods [2]. Moreover, one can distinguish between alternating [3], [4] and rotating test signal injections [5], [6]. Low- and zero-speed self-sensing control is challenging because the machine has to provide a resistive or inductance saliency in the desired current operating range [7]. This is mainly a machine design problem. Thus, a lot of research is on the optimization of machines for self-sensing control feasibility, e.g., [8] and [9]. Moreover, the control performance suffers from secondary saliencies such as cross saturation [10] and higher harmonic inductance saliencies [11], [12]. Recent papers [13] highlight that secondary resistive saliencies can also deteriorate voltage injection-based self-sensing control performance, and thus, their compensation might be necessary [14].

Test current injection is still unusual. A comparison of test voltage and test current injection has been already published in 1998 [15]. However, the focus of research was on voltage injection methods, which evaluate the current response in order to yield a position estimate. For test current injection, an extended control scheme is needed in order to provide zero steady-state control error in case of superimposed sinusoidal HF reference signals. Using test current injection, proper separation of the HF and fundamental components is provided by the controller itself in case of zero or near-zero control error. In transient operation, however, interference of fundamental and HF components might occur, which deteriorates self-sensing position estimation. To overcome this limitation, the two-degree-of-freedom current control scheme proposed in [16] is used. The (fundamental frequency) reference response is set by a model-based dynamic feedforward control, which has been

already successfully used for compensation of higher harmonic currents in PMSMs [17]. A characteristic of two-degree-of-freedom control is that the reference response is set by the dynamic feedforward control, whereas the tracking controllers are just needed for disturbance rejection. In this way, one can overcome the problem of bandwidth limitation of test current injection-based self-sensing, which existed so far. In case of self-sensing control, the tracking controllers do also provide the position error signal to drive a position and speed observer. The fundamental and HF components are also decoupled in case of transient operation. Thus, a high-performance position error signal is provided without the need for additional filtering. The innovation in this paper is that the PI+resonant controller used in [16] is modified that it directly provides a dc error signal without the need for further demodulation and delay compensation (due to the discrete-time implementation of the control algorithm). Moreover, it is analytically shown that, in contrast to voltage injection-based methods, only precise knowledge of the inductance $m_{qd}^{\text{HF}}(i_d, i_q, \gamma)$ is needed for compensation of secondary saliencies in the case of inductance saliency-based self-sensing control and $r_{qd}^{\text{HF}}(i_d, i_q, \gamma)$ in the resistance-based case, respectively. In order to further enhance control performance, a flatness-based test signal precontrol, which inherently includes compensation for secondary saliencies, is used, as proposed in [16]. The error signal for self-sensing operation is provided by evaluation of the HF actuating signals of the tracking controller. A speed estimate, by evaluating the back electromotive force (EMF) information of the machine, is used to improve dynamic behavior [16]. The effectiveness of the proposed novel self-sensing control method is validated by test bench measurement results for an interior PMSM (IPMSM), which shows strong saturation effects and higher harmonic saliencies.

This paper is organized as follows. The basis for model-based control design is a sufficiently precise model of the machine. Therefore, in Section II, a fundamental and an HF model of the PMSM are introduced. In Section III, the principle of alternating test current injection-based self-sensing control is explained, whereas Section IV addresses the two-degree-of-freedom current control scheme. In Section V, speed and position estimation is dealt with. In Section VI, experimental results give proof of the proper functioning of the proposed method.

II. PMSM MODEL

Superimposing a test signal, whose frequency is significantly higher than the fundamental frequency of the machine, requires separate modeling of the fundamental and test signal frequency behavior of the machine. To do so, the actuating signals and currents are split up into fundamental (f) and HF (HF) components, whereby $\underline{X}_{dq} = [X_d, X_q]^T$ represents voltages or currents in field-oriented coordinates

$$\underline{X}_{dq} = \underline{X}_{dq}^f + \underline{X}_{dq}^{\text{HF}}. \quad (1)$$

A. Nonideal Fundamental Frequency Model Considering Secondary Inductance Saliencies

In practice, machines typically do not only feature one single sinusoidal saliency. Secondary saliencies may occur due to the

machine geometry, like stator slotting, or due to saturation effects. To account for this, the extended machine model presented in [18] is used

$$\begin{bmatrix} u_d^f \\ u_q^f \end{bmatrix} = \underline{R}_{dq} \begin{bmatrix} i_d^f \\ i_q^f \end{bmatrix} + \underbrace{\begin{bmatrix} l_{dd} & m_{dq} \\ m_{qd} & l_{qq} \end{bmatrix}}_{\underline{L}_{dq}} \frac{d}{dt} \begin{bmatrix} i_d^f \\ i_q^f \end{bmatrix} + \omega_{\text{el}} \underbrace{\begin{bmatrix} \Psi_d^\gamma \\ \Psi_q^\gamma \end{bmatrix}}_{\underline{\Psi}_{dq}^\gamma}. \quad (2)$$

$\underline{R}_{dq} = \text{diag}(R)$ is the diagonal resistance matrix. The matrix of the differential inductances $\underline{L}_{dq}$ consists of differential self-inductances l_{dd} and l_{qq} and differential mutual inductances $m_{dq} = m_{qd}$ [19], which can occur due to cross saturation and higher harmonic components in the inductances. Ψ_x^γ denotes a change of flux with the rotor position γ , which causes an induced voltage in the machine axis $x \in \{d, q\}$ and depends on the PM flux and differential and absolute inductances [18]. In case of nonlinear material characteristics, one cannot distinguish whether a contribution to the flux is caused by the d - or q -axis current. To account for this, the new parameters Ψ_x^γ were defined in [18]. All inductance and flux parameters are defined to be current and position dependent.

B. Extension of the Nonideal Model for HF-Related Resistive Components (HF Model)

Recent papers show that HF resistive saliencies also can be used for self-sensing [20]. The underlying physical effect is that additional ac losses occur in case of HF injection, which depend on the currents and the rotor position. To deal with this phenomenon, additional ac resistances are introduced. Because test signal injection-based self-sensing control is just used for low- and zero-speed operation, the motional EMF is neglected by setting $\omega_{\text{el}} = 0$. The model proposed in [21] is used in this paper

$$\begin{bmatrix} u_d^{\text{HF}} \\ u_q^{\text{HF}} \end{bmatrix} = \underbrace{\begin{bmatrix} r_{dd}^{\text{HF}} & r_{dq}^{\text{HF}} \\ r_{qd}^{\text{HF}} & r_{qq}^{\text{HF}} \end{bmatrix}}_{\underline{R}_{dq}^{\text{HF}}} \begin{bmatrix} i_d^{\text{HF}} \\ i_q^{\text{HF}} \end{bmatrix} + \underbrace{\begin{bmatrix} l_{dd}^{\text{HF}} & m_{dq}^{\text{HF}} \\ m_{qd}^{\text{HF}} & l_{qq}^{\text{HF}} \end{bmatrix}}_{\underline{L}_{dq}^{\text{HF}}} \frac{d}{dt} \begin{bmatrix} i_d^{\text{HF}} \\ i_q^{\text{HF}} \end{bmatrix}. \quad (3)$$

All parameters are defined to be position and current dependent. Therefore, the model includes all kinds of saliencies that may occur in a PMSM. In general, the inductance and resistance parameters depend on the injection frequency and amplitude, e.g., [21]. To indicate the frequency for which the parameters are valid, superscript (HF) was introduced. The experiments carried out in [21] indicated the existence of secondary diagonal elements r_{dq}^{HF} and r_{qd}^{HF} in the resistance matrix $\underline{R}_{dq}^{\text{HF}}$.

III. PRINCIPLE OF ALTERNATING TEST CURRENT INJECTION-BASED SELF-SENSING CONTROL FOR LOW-SPEED RANGE

There are mainly two basic ways to perform alternating test current injection-based self-sensing control, i.e., HF resistance and inductance saliency-based self-sensing control. To gain the error signals for both ways, model (3) is used, which includes

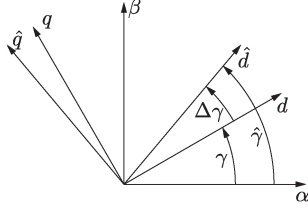


Fig. 1. Coordinate systems. The orthogonal axes of the stator fixed two-phase equivalent coordinates are denoted by α and β . The field-oriented coordinate system consists of the orthogonal d - and q -axes, whereby the angle of the d -axis with respect to the α -axis is given by γ . The estimated field-oriented coordinate system, which is misaligned by the angle $\Delta\gamma = \hat{\gamma} - \gamma$ are given by $\hat{d}$ and $\hat{q}$.

all kinds of saliencies by use of current- and position-dependent parameters.

A. HF Model in Estimated $\hat{d}\hat{q}$ Coordinates

To derive a position error signal, the system description (3) is transformed to the estimated $\hat{d}\hat{q}$ coordinate system, which is misaligned by the angle $\Delta\gamma = \hat{\gamma} - \gamma$, as shown in Fig. 1. The coordinate transformation from the field oriented into the estimated coordinates and its inverse are given by

$$\underline{X}_{dq} = \underline{T}_{\Delta\gamma} \underline{X}_{\hat{d}\hat{q}} \quad (4)$$

$$\underline{X}_{\hat{d}\hat{q}} = \underline{T}_{\Delta\gamma}^{-1} \underline{X}_{dq} \quad (5)$$

$$\underline{T}_{\Delta\gamma} = \begin{bmatrix} \cos(\Delta\gamma) & -\sin(\Delta\gamma) \\ \sin(\Delta\gamma) & \cos(\Delta\gamma) \end{bmatrix} \quad (6)$$

$$\underline{T}_{\Delta\gamma}^{-1} = \underline{T}_{\Delta\gamma}^T. \quad (7)$$

The time derivative of (6) is

$$\frac{d}{dt} \underline{T}_{\Delta\gamma} = \Delta\dot{\gamma} \underbrace{\begin{bmatrix} -\sin(\Delta\gamma) & -\cos(\Delta\gamma) \\ \cos(\Delta\gamma) & -\sin(\Delta\gamma) \end{bmatrix}}_{\underline{T}_{\Delta\gamma}^{\Delta\gamma}}. \quad (8)$$

Transforming (3) into the estimated coordinates using (4)–(8) results in

$$\underline{u}_{\hat{d}\hat{q}}^{\text{HF}} = \underline{R}_{\hat{d}\hat{q}}^{\text{HF}} \underline{i}_{\hat{d}\hat{q}}^{\text{HF}} + \Delta\dot{\gamma} \underline{L}_{\hat{d}\hat{q}}^{\text{HF}, \Delta\gamma} \underline{i}_{\hat{d}\hat{q}}^{\text{HF}} + \underline{L}_{\hat{d}\hat{q}}^{\text{HF}} \frac{d}{dt} \underline{i}_{\hat{d}\hat{q}}^{\text{HF}} \quad (9)$$

$$\underline{R}_{\hat{d}\hat{q}}^{\text{HF}} = \underline{T}_{\Delta\gamma}^{-1} \underline{R}_{dq}^{\text{HF}} \underline{T}_{\Delta\gamma} \quad (10)$$

$$\underline{L}_{\hat{d}\hat{q}}^{\text{HF}, \Delta\gamma} = \underline{T}_{\Delta\gamma}^{-1} \underline{L}_{dq}^{\text{HF}} \underline{T}_{\Delta\gamma} \quad (11)$$

$$\underline{L}_{\hat{d}\hat{q}}^{\text{HF}} = \underline{T}_{\Delta\gamma}^{-1} \underline{L}_{dq}^{\text{HF}} \underline{T}_{\Delta\gamma}. \quad (12)$$

To simplify further calculations, the following expressions are defined:

$$r_m = \frac{r_{dd}^{\text{HF}} + r_{qq}^{\text{HF}}}{2} \quad \Delta r = \frac{r_{dd}^{\text{HF}} - r_{qq}^{\text{HF}}}{2} \quad (13)$$

$$l_m = \frac{l_{dd}^{\text{HF}} + l_{qq}^{\text{HF}}}{2} \quad \Delta l = \frac{l_{dd}^{\text{HF}} - l_{qq}^{\text{HF}}}{2}. \quad (14)$$

This way, the parameter matrices as to (10)–(12) result as given in (15)–(17), shown at the bottom of the page, whereby $r_{dq}^{\text{HF}} = r_{qd}^{\text{HF}}$ as to [22].

B. Error Signals for Alternating Injection in the $\hat{d}$ -Axis

Without loss of generality, a cosine-shaped alternating test current is chosen to be injected into the estimated $\hat{d}$ -axis

$$\hat{i}_d^{\text{HF}} = \hat{I}_{\text{HF}} \cos(\omega_{\text{HF}} t) \quad (18)$$

$$\frac{d}{dt} \hat{i}_d^{\text{HF}} = -\omega_{\text{HF}} \hat{I}_{\text{HF}} \sin(\omega_{\text{HF}} t). \quad (19)$$

The HF component of the estimated $\hat{q}$ -axis current is controlled to zero by a suitable controller. For injection in the estimated $\hat{d}$ -axis, the $\hat{q}$ -axis voltage component $u_{\hat{q}}^{\text{HF}}$ has to be evaluated.

Splitting up of $u_{\hat{q}}^{\text{HF}}$ for sine and cosine parts provides

$$u_{\hat{q}}^{\text{HF}} = u_{\hat{q}, \cos}^{\text{HF}} \cos(\omega_{\text{HF}} t) + u_{\hat{q}, \sin}^{\text{HF}} \sin(\omega_{\text{HF}} t). \quad (20)$$

Evaluating (9), one yields for the sine and cosine coefficients

$$u_{\hat{q}, \cos}^{\text{HF}} = (-\Delta r \sin(2\Delta\gamma) + r_{dq}^{\text{HF}} \cos(2\Delta\gamma) + \Delta\dot{\gamma} (l_m - \Delta l \cos(2\Delta\gamma) - m_{qd}^{\text{HF}} \sin(2\Delta\gamma))) \hat{I}_{\text{HF}} \quad (21)$$

$$u_{\hat{q}, \sin}^{\text{HF}} = (\Delta l \sin(2\Delta\gamma) - m_{qd}^{\text{HF}} \cos(2\Delta\gamma)) \omega_{\text{HF}} \hat{I}_{\text{HF}}. \quad (22)$$

Analyzing (21) and (22), one finds the following characteristics.

- The main resistance saliency Δr , which is to be tracked in the case of resistance-based self-sensing control, occurs in the cosine component of the HF actuating signal in the $\hat{q}$ -axis.
- The main inductance saliency Δl , which is to be tracked in the case of inductance-based self-sensing control, occurs in the sine component of the HF actuating signal in the $\hat{q}$ -axis.
- In case no secondary resistive saliencies occur ($r_{dq}^{\text{HF}} = r_{qd}^{\text{HF}} = 0$), driving the HF voltage $u_{\hat{q}, \cos}^{\text{HF}}$ to zero provides

$$\underline{R}_{\hat{d}\hat{q}}^{\text{HF}} = \begin{bmatrix} r_m + \Delta r \cos(2\Delta\gamma) + r_{dq}^{\text{HF}} \sin(2\Delta\gamma) & -\Delta r \sin(2\Delta\gamma) + r_{dq}^{\text{HF}} \cos(2\Delta\gamma) \\ -\Delta r \sin(2\Delta\gamma) + r_{dq}^{\text{HF}} \sin(2\Delta\gamma) & r_m - \Delta r \cos(2\Delta\gamma) - r_{dq}^{\text{HF}} \sin(2\Delta\gamma) \end{bmatrix} \quad (15)$$

$$\underline{L}_{\hat{d}\hat{q}}^{\text{HF}, \Delta\gamma} = \begin{bmatrix} -\Delta l \sin(2\Delta\gamma) + m_{dq}^{\text{HF}} \cos(2\Delta\gamma) & -l_m - \Delta l \cos(2\Delta\gamma) - m_{dq}^{\text{HF}} \sin(2\Delta\gamma) \\ l_m - \Delta l \cos(2\Delta\gamma) - m_{dq}^{\text{HF}} \sin(2\Delta\gamma) & \Delta l \sin(2\Delta\gamma) - m_{dq}^{\text{HF}} \cos(2\Delta\gamma) \end{bmatrix} \quad (16)$$

$$\underline{L}_{\hat{d}\hat{q}}^{\text{HF}} = \begin{bmatrix} l_m + \Delta l \cos(2\Delta\gamma) + m_{dq}^{\text{HF}} \sin(2\Delta\gamma) & -\Delta l \sin(2\Delta\gamma) + m_{dq}^{\text{HF}} \cos(2\Delta\gamma) \\ -\Delta l \sin(2\Delta\gamma) + m_{dq}^{\text{HF}} \sin(2\Delta\gamma) & l_m - \Delta l \cos(2\Delta\gamma) - m_{dq}^{\text{HF}} \sin(2\Delta\gamma) \end{bmatrix} \quad (17)$$

zero steady-state position estimation error, as can be seen from the corresponding error signal

$$\Delta\gamma = \frac{1}{2} \arcsin \left(\frac{u_{\hat{q},\cos}^{\text{HF}}}{-\Delta r \hat{I}_{\text{HF}}} \right). \quad (23)$$

- In case no secondary inductance saliencies occur ($m_{dq}^{\text{HF}} = m_{qd}^{\text{HF}} = 0$), driving the HF voltage $u_{\hat{q},\sin}^{\text{HF}}$ to zero provides zero steady-state position estimation error, as can be seen from the corresponding error signal

$$\Delta\gamma = \frac{1}{2} \arcsin \left(\frac{u_{\hat{q},\sin}^{\text{HF}}}{\Delta l \omega_{\text{HF}} \hat{I}_{\text{HF}}} \right). \quad (24)$$

- In case secondary resistive or inductance saliencies occur, a compensation voltage has to be provided to compensate for the position estimation error, which would occur otherwise

$$\Delta\gamma_{\text{err,resist}} = \frac{1}{2} \arctan \left(\frac{2r_{dq}^{\text{HF}}}{r_{dd}^{\text{HF}} - r_{qq}^{\text{HF}}} \right) \quad (25)$$

$$\Delta\gamma_{\text{err,induct}} = \frac{1}{2} \arctan \left(\frac{2m_{dq}^{\text{HF}}}{l_{dd}^{\text{HF}} - l_{qq}^{\text{HF}}} \right). \quad (26)$$

One can see in (25) that, in the case of resistance saliency-based self-sensing control, the steady-state error does just depend on resistance parameters. In inductance-based self-sensing control, the steady-state error (26) just depends on inductance parameters. This is a crucial advantage of alternating test current injection-based self-sensing control because no secondary resistive saliencies have to be decoupled in the case of inductance saliency-based self-sensing control and vice versa.

- It is worth mentioning that, in the case of resistive saliency-based self-sensing control, an additional disturbance occurs in (21), which is just vanishing for $\Delta\gamma = 0$.

The position error signals are ambiguous; thus, an initialization process is needed for initial start-up [23]. Moreover, there is a limit for the maximum position error that may occur during operation; otherwise, a 180° phase shift in the estimated position might occur, causing instability of the control. The position error information gets lost if $\Delta r = 0$ or $\Delta l = 0$.

IV. TWO-DEGREE-OF-FREEDOM CURRENT CONTROL SCHEME

The self-sensing current control scheme has to fulfill various requirements.

- The HF signal should not affect the fundamental current reference response and vice versa (proper separation).
- Provide zero steady-state control error for fundamental and HF current reference values/trajectories.
- Enable demodulation of the HF sine or cosine part of the $\hat{q}$ -axis control signal.

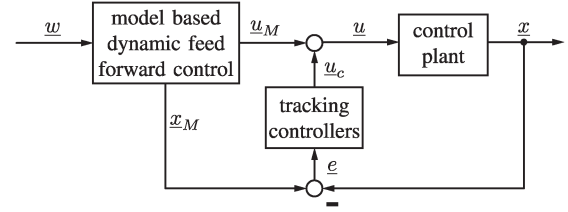


Fig. 2. Two-degree-of-freedom control scheme consisting of a model-based dynamic feedforward control, which generates control signals $\underline{u}_M$ from the reference values $\underline{w}$. The model state vector $\underline{x}_M$ is used as a reference value for the tracking controllers, which generate the control signals $\underline{u}_c$ based on the control error $\underline{e} = \underline{x}_M - \underline{x}$.

- Compensate for nonideal characteristics of the machine such as secondary saliencies.

To fulfill these requirements, a two-degree-of-freedom control scheme, as depicted in Fig. 2, is used, which features the following four properties.

- Two-degree-of-freedom control is a model-based control scheme. That is, a model of the control plant is included in the control scheme.
- The reference response and disturbance rejection can be set independently from each other.
- The reference response is determined by the model-based dynamic feedforward control, e.g., by means of suitable eigenvalue assignment.
- Modeling errors and disturbances are rejected by the (model reference) tracking controllers. Suitable internal models are needed to ensure zero steady-state control error.

The idea of model reference tracking control is to provide the major part of the actuating signal by the feedforward control even during transients. In this way, proper separation of fundamental and HF components is also ensured in dynamic operation. The model currents are used as a reference for the tracking controllers. The tracking controllers provide zero steady-state control error if there are model parameter errors. In case of self-sensing control, they also do provide the position error signal, which is used to drive a position and speed observer, as depicted in Fig. 3. A compensation voltage for position estimation errors caused by secondary saliencies is provided by a flatness-based test signal precontrol.

A. Feedforward Control of (Fundamental) Reference Response

The fundamental frequency behavior of the machine is modeled by (2). This model is used to derive the model-based dynamic feedforward control for the fundamental currents

$$\frac{d}{dt} \underline{i}_{dq}^f = -\underline{L}_{dq}^{-1} \underline{R}_{dq} \underline{i}_{dq}^f + \underline{L}_{dq}^{-1} \underline{u}_{dq}^f - \omega_{el} \underline{L}_{dq}^{-1} \underline{\Psi}_{dq}^\gamma. \quad (27)$$

Due to low-speed operation, the back EMF is neglected by setting $\omega_{el} = 0$. Using Falb–Wolovich decoupling theory for linear systems [24] provides the decoupling control law

$$\underbrace{\begin{bmatrix} u_d^f \\ u_q^f \end{bmatrix}}_{\underline{u}_{dq}^f} = \underbrace{\begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix}}_{\underline{R}_{dq}} \underbrace{\begin{bmatrix} i_d^f \\ i_q^f \end{bmatrix}}_{\underline{i}_{dq}^f} + \underbrace{\begin{bmatrix} l_{dd} & m_{dq} \\ m_{qd} & l_{qq} \end{bmatrix}}_{\underline{L}_{dq}} \begin{bmatrix} v_d \\ v_q \end{bmatrix}. \quad (28)$$

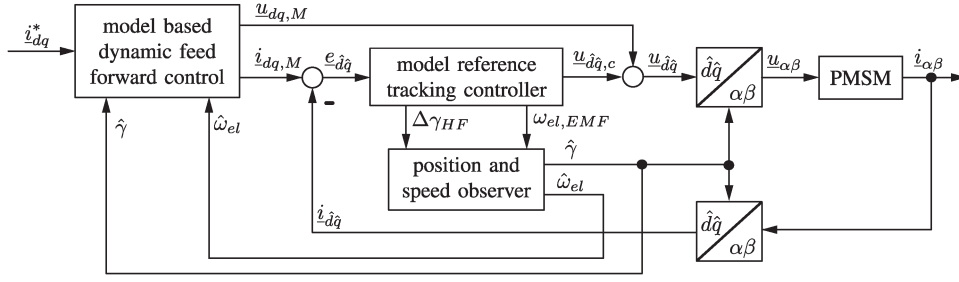


Fig. 3. Two-degree-of-freedom current control scheme, including self-sensing position and speed estimation using a position error signal and a speed estimate obtained from the tracking controller.

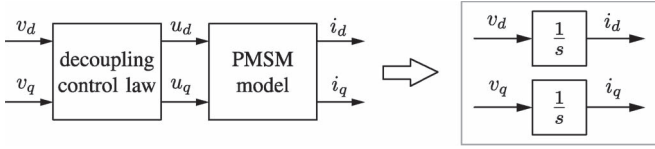


Fig. 4. Principle of Falb-Wolovich decoupling control providing a linear state-space description, which just consists of integrators (gray box).

The decoupled system description using (28) just consists of two integrators (see Fig. 4), whereby the new system input $\underline{v}_{dq} = [v_d \ v_q]^T$ is defined as

$$\underline{v}_{dq} = \frac{d}{dt} \underline{i}_{dq}^f. \quad (29)$$

The system description in new coordinates is unstable and thus needs to be stabilized using constant state feedback $\underline{k}_{dq} = [k_d \ k_q]$, as depicted in Fig. 5. Constant state feedback is sufficient because the integrators force constant errors (reference values are dc signals in steady state) $e_{dq} = i_{dq}^* - i_{dq,M}$ to zero. The desired reference response can be set by suitable choice of k_d and k_q , e.g., using eigenvalue assignment. It is worth pointing out that the feedforward control does not affect stability of the control system.

B. Flatness-Based Test Signal Precontrol

In order to obtain a position error signal for low- and zero-speed operation, an HF cosine-shaped test current is superimposed to the fundamental current. The feedforward control presented in the previous section is not able to provide zero steady-state control error in case of superimposing an alternating test current reference. In contrast to the fundamental reference values, the test current reference trajectory is known beforehand in an analytical way as to (18). Using the concept of differential flatness [25], the test current precontrol can be found in a very elegant way [16]. The HF model (3), which represents the behavior of the machine in case of injecting an HF test signal, is used to derive the test signal precontrol.

The test signal precontrol law is an algebraic calculation formula for the HF actuating signal trajectory based on the knowledge of the test signal amplitude $\hat{I}_{HF}$ and frequency f_{HF} . The test current is chosen to be injected into the $\hat{d}$ -axis as to (18), whereas the $\hat{q}$ -axis current is kept zero, resulting in

$$\begin{bmatrix} u_{d, HF} \\ u_{q, HF} \end{bmatrix} = \hat{I}_{HF} \begin{bmatrix} r_{dd}^{HF} \cos(\omega_{HF} t) - \omega_{HF} l_{dd}^{HF} \sin(\omega_{HF} t) \\ r_{qd}^{HF} \cos(\omega_{HF} t) - \omega_{HF} m_{qd}^{HF} \sin(\omega_{HF} t) \end{bmatrix}. \quad (30)$$

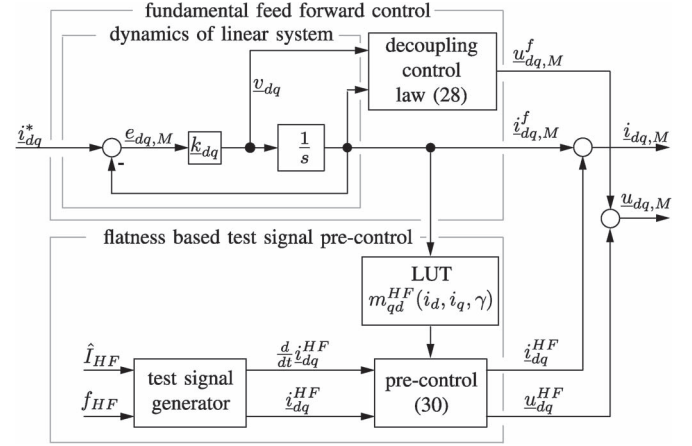


Fig. 5. Model-based dynamic feedforward control consisting of a model-based dynamic feedforward control for fundamental current and flatness-based test signal precontrol. Dynamics of the fundamental feedforward control are set by suitable choice of the state feedback $\underline{k}_{dq} = [k_d \ k_q]$. The reference value from the speed controller or the torque controller is i_{dq}^* . The model state $i_{dq,M}$ is the reference value for the tracking controller. Parameter $m_{qd}^{HF}(i_d, i_q, \gamma)$ determines the steady-state position estimation error in case of inductance saliency-based self-sensing control and therefore is stored in a lookup table (LUT). In case of resistance-based self-sensing control, $r_{qd}^{HF}(i_d, i_q, \gamma)$ is stored in the LUT instead of $m_{qd}^{HF}(i_d, i_q, \gamma)$.

The actuating signal (30) is added to the fundamental actuating signal generated by the fundamental feedforward control as to Fig. 5. In case there are no initial, model, parameter, or position estimation errors, the real system tracks the desired reference trajectory perfectly.

The compensation for secondary saliencies is inherently included in the flatness-based precontrol because the reference model (3) already includes secondary saliencies. In case of inductance saliency-based self-sensing control, the quality of the compensation just depends on the knowledge of the position-dependent parameter $m_{qd}^{HF}(i_d, i_q, \gamma)$, as shown in error equation (22). In case of resistance saliency-based self-sensing control, the relevant parameter is $r_{qd}^{HF}(i_d, i_q, \gamma)$, respectively. Both parameters most often depend more or less significantly on the electric rotor position. Because the real electric rotor position is not available from a sensor measurement, the estimated position $\hat{\gamma}$ from the position and speed observer is fed to the LUT in the feedforward. This does not affect stability of the control system because the flatness-based precontrol does just provide a compensation voltage in a feedforward manner, although the position estimate from the observer is used. As long as the parameter in the LUT matches the real parameter, no ripple

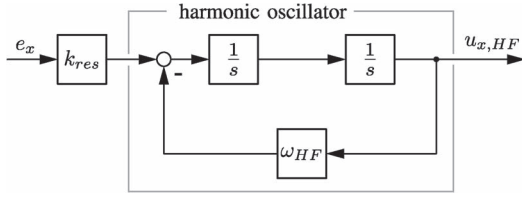


Fig. 6. Conventional resonant controller consisting of a harmonic oscillator and gain parameter k_{res} to set the desired dynamics. The input is the control error e_x ($x \in \{d, q\}$), and $u_{x,HF}$ is the actuating signal from the resonant controller.

(oscillations due to harmonics) or offset (due to errors in the average value) occurs in the position estimate. With increasing parameter mismatch, also an increasing ripple in the position estimate is observed by reason of imperfect compensation. Just either m_{qd}^{HF} or r_{qd}^{HF} are stored as current- and position-dependent parameters. For the other parameters, just constants are used to reduce memory requirements. This does not affect steady-state precision of self-sensing control.

Due to discrete-time implementation of the control scheme, the feedforward control uses $\gamma_{HF}[k+1] = \omega_{HF}t[k+1]$ to account for the dead time. Like this, the delay due to the hold element and computation time, which is assumed to be half the PWM period, is compensated for.

C. Model Reference Tracking Controllers

If the model, its parameters, and the initial conditions were perfectly known, the model-based feedforward control would provide perfect actuating signals, which provide zero tracking error. In practice, however, model errors and a misalignment of the estimated rotor angle will cause control errors, which have to be compensated for by use of suitable tracking controllers.

The internal model principle is fundamental to prove whether a control system is able to provide zero steady-state control error for a certain kind of reference or disturbance signal. This principle provides that the controller or the control plant itself must contain a signal generator that has the poles or eigenvalues of the reference or disturbance signal [26]. The conventional PI-type controllers in field-oriented coordinates are not able to provide zero steady-state control error in case of injecting sinusoidal HF test currents. Therefore, an additional controller in parallel to the fundamental frequency controller is needed [15], [27], [28]. A resonant controller, as for example the one used in [29], includes a harmonic oscillator (signal generator) with eigenfrequency $\omega_{res} = \omega_{HF}$, thus fulfilling the internal model principle and is given as

$$F_{res}(s) = k_{res} \cdot \frac{s}{s^2 + \omega_{HF}^2}. \quad (31)$$

The block diagram of the resonant controller (31) is depicted in Fig. 6 and provides ac actuating signals $u_{x,HF}$ with frequency ω_{HF} . However, for self-sensing control, a dc error signal is desired. Therefore, quadrature amplitude demodulation and compensation for delay times caused by the discrete-time implementation of the control was used in [16]. Delay-time compensation is necessary to be able to guarantee separation of the resistive and inductance components, which is a key feature

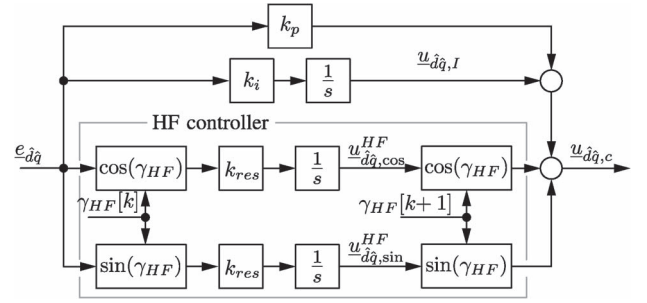


Fig. 7. Model reference tracking controller consisting of the fundamental PI-type controller and the HF controller in parallel.

of alternating test current injection. If, for example, the test signal frequency was $f_{HF} = (1/8)f_s$, whereby f_s is the sampling frequency of the current controller, a phase error of 22.5° would occur in the actuating signals at test signal frequency just by reason of the hold element, which is caused by discrete-time implementation. Therefore, one can conclude that delay compensation is mandatory. Moreover, the controller provides separation of the fundamental frequency and HF components. Therefore, in contrast to voltage injection-based methods, no additional filters for separation of these components are needed.

The innovation in this paper is the HF controller depicted in Fig. 7, which has the input/output behavior of a resonant controller, including delay-time compensation (for one zero-order hold element and control algorithm computation time $T_{PWM}/2$) by using $\gamma_{HF}[k+1]$ instead of $\gamma_{HF}[k]$ for the back transformation [30]. This controller directly provides dc signals and inherently includes demodulation for sine and cosine components of the HF actuating signals. Compared with Fourier series expansion, one finds that the structure contains a calculation of the sine and cosine Fourier coefficients for the HF frequency.

V. POSITION AND SPEED OBSERVER

The position error signal and a speed estimate based on back EMF evaluation are used to drive a Luenberger observer to gain a position and speed estimate needed for self-sensing field-oriented current and speed control.

A. Speed Estimation Based on Back EMF Evaluation

By setting parameter Ψ_q^γ to zero in the feedforward control, a high-quality speed estimate can be obtained from evaluating the actuating signal of the $\hat{q}$ -axis controller. In case there are no model and parameter errors, the fundamental tracking controllers just have to compensate for the induced voltage $\omega_{el}\Psi_q^\gamma$. If there is no position estimation error ($\Delta\gamma = 0$), the fundamental actuating signals of the controllers are

$$\begin{bmatrix} u_{d,c} \\ u_{q,c} \end{bmatrix} = \omega_{el} \begin{bmatrix} 0 \\ \Psi_q^\gamma \end{bmatrix}. \quad (32)$$

This leads to the speed estimate

$$\hat{\omega}_{el,EMF} \approx \frac{u_{\hat{q},c}}{\Psi_q^\gamma}. \quad (33)$$

In steady-state operation, the precision of this estimate just depends on the knowledge of parameter Ψ_q^γ . During transients, there might be some disturbance from the transient operation of the tracking controller by reason that the machine parameters implemented in the model-based dynamic feedforward are not perfectly known. However, the effects are rather small because the main part of the actuating signal for transient operation is provided by the model-based dynamic feedforward. This is a significant advantage of the two-degree-of-freedom control scheme, which is used in this paper.

Parameter Ψ_q^γ might change during operation, e.g., due to a change of magnet temperature. However, this is not a severe problem because the estimate does not need to be perfect. The main idea is to enhance the performance of the position and speed observer, which is introduced in the next section. This observer compensates for errors in the speed estimate anyway.

B. Position and Speed Estimation

A mechanical model in state-space description is the basis for observer design

$$\dot{x} = Ax + Bu \quad (34)$$

$$\dot{\gamma} = \omega_{el}. \quad (35)$$

The state variable $x = \gamma$ is the electric rotor position, and the input $u = \omega_{el}$ is the (electric) angular speed. Moreover, $A = 0$ and $B = 1$.

Due to parameter errors, the speed estimate $\hat{\omega}_{el,EMF}$, provided by evaluating the back EMF, would not match the real speed ω_{el} , resulting in steady-state speed estimation errors. To deal with this, the speed estimation error $\Delta\omega_{el} = \omega_{el} - \hat{\omega}_{el,EMF}$ is introduced and used as an additional observer state variable, resulting in the extended model

$$\dot{x}_e = A_e x_e + B_e u_e \quad (36)$$

$$\frac{d}{dt} \begin{bmatrix} \gamma \\ \Delta\omega_{el} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{A_e} \begin{bmatrix} \gamma \\ \Delta\omega_{el} \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{B_e} \hat{\omega}_{el,EMF}. \quad (37)$$

Thereby, the extended state vector is $x_e = [\gamma \ \Delta\omega_{el}]^T$. The differential equation $\Delta\omega_{el} = 0$ is a disturbance model for constant disturbances to account for errors in $\hat{\omega}_{el,EMF}$ by reason of parameter errors. Like this, zero steady-state speed estimation error is guaranteed. The measurement equation for the observer is

$$y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C \begin{bmatrix} \gamma \\ \Delta\omega_{el} \end{bmatrix}. \quad (38)$$

The Luenberger-type observer, which is simply a disturbance observer, is given by the following state-space equations:

$$\dot{\hat{x}}_e = A_e \hat{x}_e + B_e u_e + L(y - \hat{y}) \quad (39)$$

$$\hat{y} = C \hat{x}_e. \quad (40)$$

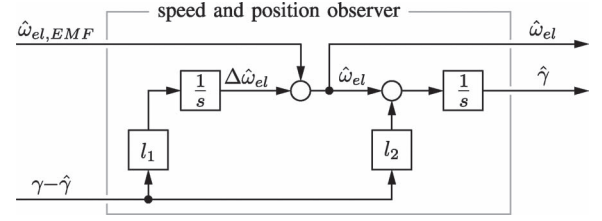


Fig. 8. Luenberger observer providing position and speed estimate based on back EMF-based speed estimate $\hat{\omega}_{el,EMF}$ and position error signal from evaluation of HF injection for low- and zero-speed operation. The desired dynamics of the observer are set by suitable choice of the feedback gains l_1 and l_2 .

Introducing the observation error $\varepsilon = x - \hat{x}$ provides the observation error dynamic, which needs to be stabilized by suitable choice of the gain vector $L = [l_1 \ l_2]^T$, e.g., using eigenvalue assignment

$$\dot{\varepsilon} = (A - LC)\varepsilon. \quad (41)$$

The resulting speed and position observer is depicted in Fig. 8. The inputs of the observer are the speed estimate $\hat{\omega}_{el,EMF}$ from evaluation of (33) using the fundamental actuating signal $u_{q,c}$ of the tracking controller and the position error signal $\Delta\gamma$ obtained from evaluation of (24) in case of inductance saliency-based self-sensing or (23) in case of resistive saliency-based self-sensing, respectively. The speed estimate from the observer can be calculated as $\hat{\omega}_{el} = \Delta\omega_{el} + \hat{\omega}_{el,EMF}$. By reason of introducing $\Delta\omega_{el}$ as additional state variable, the observer guarantees zero steady-state speed estimation error independently from any machine parameter, although the estimate of (33) might be imperfect. The steady-state position estimation error just relies on the precise knowledge of the parameter m_{qd}^{HF} in case of inductance saliency-based self-sensing or r_{qd}^{HF} in case of resistance-based self-sensing.

VI. TEST SETUP AND EXPERIMENTAL RESULTS

The experiments were carried out using an IPMSM with concentrated windings, which shows significant saturation and sixth-harmonic effects. The most important control and nominal motor parameters are given in Table I. The test signal frequency is chosen to be $f_{HF} = 1000$ Hz, and the test current amplitude is $\hat{I}_{HF} = 250$ mA. The voltage required for injecting the test current depends on the operating point and is (for the analyzed operating range) on the order of 15–25 V, which is fairly small compared with the dc-link voltage of the converter of $V_{DC} = 600$ V.

A. Test Setup

The machine was supplied by an industrial converter, whereby the PWM control signals for the half-bridges of the converter were supplied by a modular DSpace real-time prototyping system DS1005 with PWM board DS5101 with a resolution of 25 ns. The switching frequency of the converter was chosen to be $f_{PWM} = 8$ kHz, and single update modulation using space vector modulation and regular current sampling method without oversampling in the middle of the PWM period was used.

TABLE I
CONTROL AND MOTOR PARAMETERS

Feed forward control gain	$1/k_d = 1/k_q = 5ms$
Tracking controller gains	$k_p = 111$ $k_i = 18500$ $k_{res} = 50000$
Observer Eigenvalues	$\lambda_1 = -155$ $\lambda_2 = -64$
Test Signal Amplitude	$\hat{I}_{HF} = 250mA$
Test Signal Frequency	$f_{HF} = 1000Hz$
PWM frequency	$f_{PWM} = 8000Hz$
Converter DC-link voltage	$V_{DC} = 600V$
Rated Motor Current	$I_{r,RMS} = 5.6A$
Rated Torque	$M_r = 16Nm$
Rated speed	$n_r = 1500rpm$
Current control limit	$I_{lim} = 10A$
Phase resistance	$R_{DC}(\vartheta = 20^\circ C) = 1.6\Omega$
Nominal Inductances	$L_d = 10mH$ $L_q = 13mH$
PM-flux	$\Psi_{PM} = 0.28Vs$
Number of poles	$2p = 10$

The converter error compensation method accounting for interlock time, current-dependent switching characteristics, and forward voltage drops of the power electronic devices, proposed in [31], was used. If not compensated for, the nonideal characteristics of the converter mainly appeared as additional fictitious resistive saliencies [14] and thus deteriorated resistance-based self-sensing. For inductance-based self-sensing, the impact of the converter errors is negligible even if the converter error compensation is switched off because alternating test current injection provides decoupling of all resistance effects, thus also including the converter errors, which can be represented as dc and higher harmonic resistances. One can conclude from this that, in case of resistance-based self-sensing, the compensation for converter errors is expected to be of significant importance. However, the converter errors do also depend on the junction temperature, which makes it complicated to achieve high-quality compensation for the converter errors. Therefore, one has to conclude that, from an industrial point of view, test current injection-based self-sensing control using inductance saliency evaluation is preferable.

B. Identified Machine Parameters

The PMSM model considering higher harmonics [18] was parameterized using the methods proposed in [21] and [32]. The identified mean inductances are depicted in Fig. 9 and were identified using measurements for different rotor positions in steps of 15 electrical degrees. By averaging all of these measurements, the mean parameters denoted by index $(_0)$ result.

The measurements show that there is strong dependence of the inductance and resistance parameters on the fundamental currents $i_{d,f}$ and $i_{q,f}$. One can also see that the HF resistances are significantly higher than the dc resistance given in Table I. This is by reason of additional losses occurred in case of injecting a test signal with a significantly higher frequency than the fundamental frequency. One can see that there is also a difference in r_{dd}^{HF} and r_{qq}^{HF} , which is a prerequisite for resistance saliency-based self-sensing. However, for the chosen test signal amplitude, the voltages, which had to be evaluated for self-sensing, are one order of magnitude smaller than in the case of inductance saliency-based self-sensing. Therefore, only inductance saliency-based self-sensing is tested and just the resistance decoupling property is proven by the experiments.

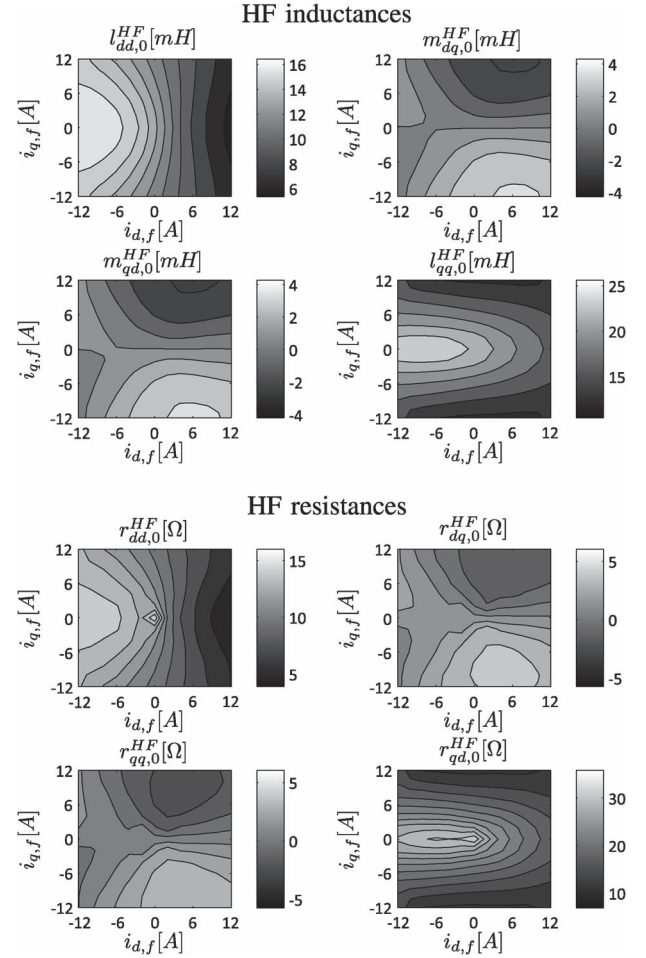


Fig. 9. Dependence of the machine parameters on the fundamental d - and q -axis currents for test signal amplitude of $\hat{I}_{HF} = 250$ mA and test signal frequency $f_{HF} = 1000$ Hz. The parameters are averaged over one electric period.

In Fig. 10, the inductance saliency without (left) and with considering higher harmonics (right) is depicted. In case of considering higher harmonics, the minimum inductance saliency over one electric period is shown because a prerequisite for self-sensing control is that the inductance saliency does not vanish for the desired current operating range in any rotor position. One can find that, for the chosen current limit of $I_{lim} = 10$ A, the machine saliency does not vanish even if harmonics are considered. However, one can see that there is a significant impact of the harmonics because zero saliency does already occur for smaller currents in the right-hand-side plot. A detailed analysis of the harmonics and their modeling is given in [18].

For inductance saliency-based self-sensing in zero- and low-speed range, precise knowledge of the parameter $m_{qd}^{HF}(i_d, i_q, \gamma)$ is needed to provide good position estimation accuracy. In this paper, it is represented as follows:

$$m_{qd}^{HF}(i_d, i_q, \gamma) = m_{qd,0}^{HF} + m_{qd,6,\cos}^{HF} \cos(6\gamma) + m_{qd,6,\sin}^{HF} \sin(6\gamma). \quad (42)$$

Parameters $m_{qd,0}^{HF}$, $m_{qd,6,\cos}^{HF}$, and $m_{qd,6,\sin}^{HF}$ for $i_d = 0$ A were identified using [21] and are depicted in Fig. 11. Although the machine shows strong saturation and harmonics, parameters l_{dd}^{HF} and l_{qq}^{HF} were implemented as constant values in the

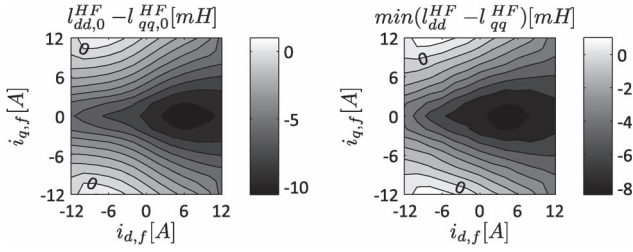


Fig. 10. Average inductance saliency $l_{dd,0}^{HF} - l_{qq,0}^{HF}$ (left) without considering higher harmonics. Minimum inductance saliency $\min(l_{dd}^{HF} - l_{qq}^{HF})$ considering higher harmonics in the inductances. Zero saliency is highlighted by a 0 in the contour plot.

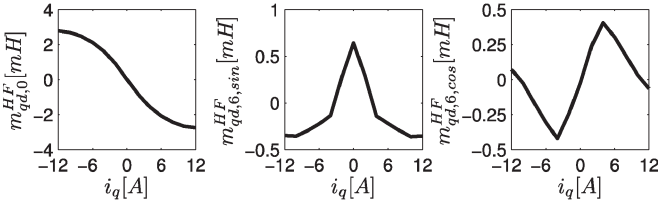


Fig. 11. Coefficients of $m_{qd}^{HF}(i_d, i_q, \gamma)$ using the representation for DC part and sixth harmonic given by (42) for $i_d = 0$ A, which needs to be known precisely; otherwise, position estimation errors would occur. These coefficients are stored in LUTs in the test signal precontrol.

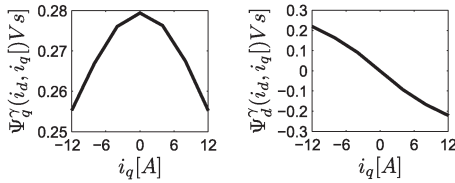


Fig. 12. Identified parameters Ψ_d^γ and Ψ_q^γ for $i_d = 0$ A, whereby Ψ_q^γ is stored in a LUT for back EMF-based speed estimation.

feedforward control. The identified flux parameters Ψ_d^γ and Ψ_q^γ for $i_d = 0$ A are given in Fig. 12, whereby Ψ_q^γ is used to gain a speed estimate based on back EMF evaluation.

C. Experimental Results for Self-Sensing Control

A comparison of the speed step response from standstill to 50 r/min with and without compensation for m_{qd}^{HF} is depicted in Fig. 13. If the compensation for m_{qd}^{HF} is used, a position estimation error of significantly less than 5° electric rotor position and a very smooth speed estimate can be observed. Otherwise, a significant sixth harmonic occurs in the position estimation error and in the estimated speed of the machine. This can be also observed in the voltage component $u_{q,\sin}^{HF}$, which is related to the position error and should be driven to zero for self-sensing control.

The tested machine shows significant saturation effects, resulting in steady-state position estimation errors, if not compensated for. In Fig. 14 a load step from zero to 90% rated torque at standstill is depicted with and without compensation for the current- and position-dependent inductance m_{qd}^{HF} . One can see that the compensation leads to a very small position estimation error, whereas without compensation, a steady-state position estimation error of nearly 20 electrical degrees occurs.

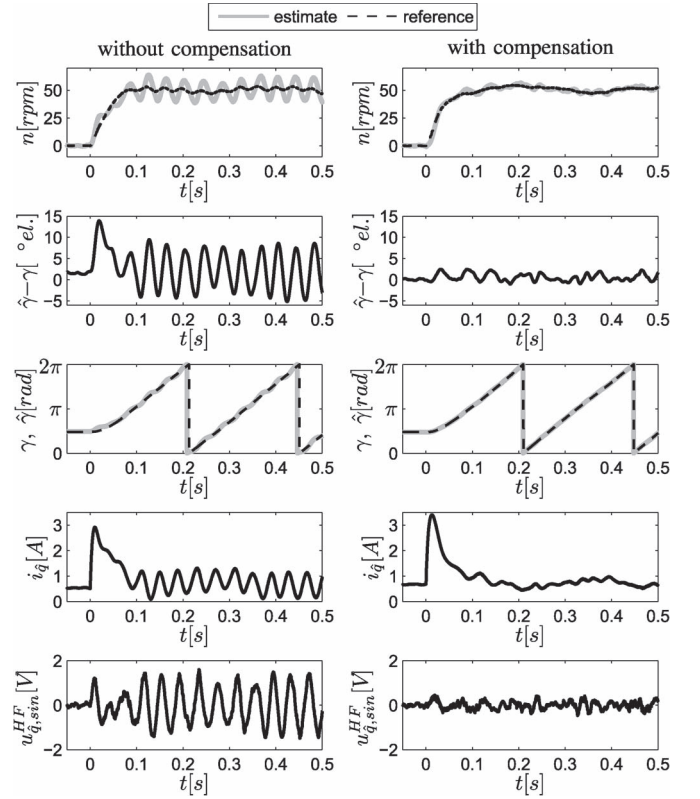


Fig. 13. Speed step response for self-sensing speed control from standstill to 50 r/min using HF test current injection (left) without and (right) with compensation for m_{qd}^{HF} . The reference torque of the load machine was set to zero. The $\hat{q}$ -axis current is nonzero in steady-state operation because torque is required to compensate for the friction torque.

No harmonics occur in steady state because the speed reference is zero in this case. Although a position estimation error of approximately 20 electrical degrees occurs without compensation for m_{qd}^{HF} , the estimated $\hat{q}$ -axis current remains nearly the same. At first sight, this is not expected for an IPMSM. Further analysis showed that the torque producing flux does not change for both cases, which nearly results in the same current level. In case of misalignment, the missing torque component from the PM flux is compensated for by the reluctance torque component of the machine.

D. Discussion of the Results

The steady-state accuracy of the proposed method just relies on precise knowledge of m_{qd}^{HF} (inductance saliency-based self-sensing) or r_{qd}^{HF} (resistance saliency-based self-sensing). However, as previously discussed, in case of resistance saliency-based self-sensing, the voltage errors caused by the nonideal characteristics of the converter have to be compensated for precisely. An important issue to discuss is the temperature dependence of these parameters. The HF resistive parameters and converter error parameters show stronger dependence on temperature than the HF inductances. Therefore, temperature effects are expected to be smaller for inductance saliency-based self-sensing. During the experiments, using inductance saliency-based self-sensing, no significant temperature dependence of the performance was observed.

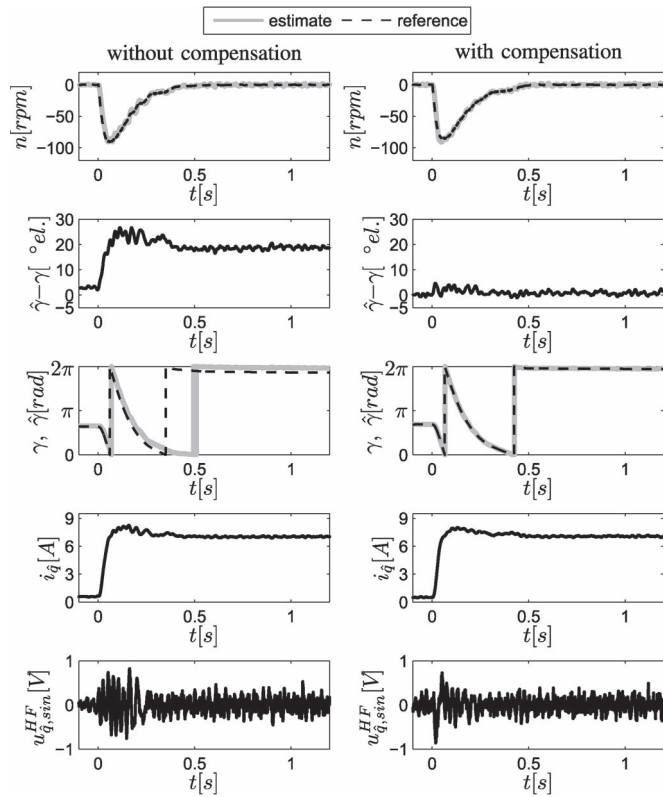


Fig. 14. Load step response from zero to 90% rated torque for standstill operation (left) without and (right) with compensation for m_{qd}^{HF} .

The machine under test was an IPMSM. That kind of machine is typically capable of providing a significant part of the torque from the magnetic reluctance by choosing negative d -axis currents in case of flux-weakening-type IPMSM ($L_d < L_q$) or by a positive d -axis current in case of flux-intensifying-type IPMSM ($L_d > L_q$) [13]. The proposed method was demonstrated in this paper for $i_d = 0$ A control strategy only. However, there is no limitation to that kind of control strategy. Using optimized control strategies such as maximum torque per ampere (MTPA) control just requires replacing the parameters identified for $i_d = 0$ A (see Figs. 11 and 12) by parameters along the MTPA trajectory and an additional current reference generator from the torque command. Successful implementation of this method was demonstrated in [33].

VII. CONCLUSION

In this paper, an innovative two-degree-of-freedom current control scheme for self-sensing control has been proposed. It is shown that, using two-degree-of-freedom control, the decoupling problem (proper separation) of fundamental reference response and HF test signal can be solved, thus providing very dynamic self-sensing operation. A complete theory for alternating test current injection for both inductance and resistance saliency-based self-sensing is provided. It was shown that, compared with inductance saliency-based self-sensing, in the case of resistance saliency-based self-sensing, an additional disturbance occurs in the error signal if the position estimation error changes. Alternating test current injection provides very

attractive features. It is analytically shown that, in the case of inductance saliency-based self-sensing control, the accuracy of position estimation just depends on the knowledge of the position- and current-dependent inductance m_{qd}^{HF} , whereas for HF resistance saliency-based self-sensing control, the important parameter is r_{qd}^{HF} . In contrast to voltage injection-based methods, the compensation for secondary saliencies is systematically included in the flatness-based test signal precontrol. The proposed tracking controller features direct demodulation of the HF sine and cosine components needed for self-sensing control, including delay-time compensation, which is mandatory to ensure proper separation of resistance and inductance effects. Two-degree-of-freedom test current injection-based control is a very systematic approach and eliminates the bandwidth limitation problem, which so far prevented a broad acceptance of test current injection-based self-sensing control. The proposed method is also applicable for optimized control strategies such as MTPA control, which is typically used for machines that are capable of providing significant reluctance torque.

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